

# 10/100BASE-T SINGLE PORT SURFACE MOUNT MAGNETICS With Various Turns Ratios



- RoHS peak reflow temperature rating: 245°C
- Meets IEEE 802.3 specification
- Compact footprint for use in the most space-restrictive applications

## Electrical Specifications @ 25°C — Operating Temperature 0°C to 70°C

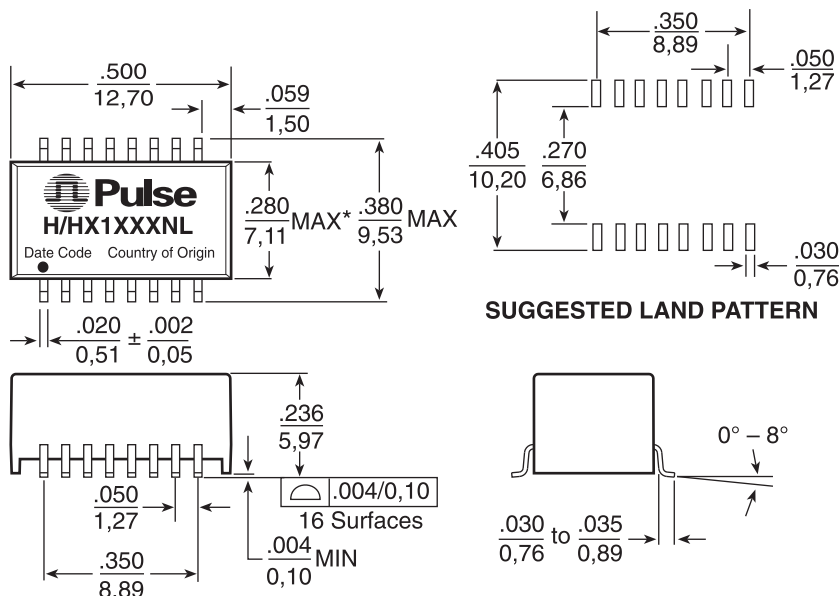
| RoHS<br>Compliant<br>Part Number | Turns Ratio |     | Insertion Loss<br>100 kHz-100 MHz<br>(dB MAX) | Return Loss<br>(dB TYP) |        |        | Differential to Common<br>Mode Rejection<br>(dB TYP) |        |         | Crosstalk<br>(dB TYP) |        |         | Hipot<br>(Vrms MIN) |
|----------------------------------|-------------|-----|-----------------------------------------------|-------------------------|--------|--------|------------------------------------------------------|--------|---------|-----------------------|--------|---------|---------------------|
|                                  | TX          | RX  |                                               | 30 MHz                  | 60 MHz | 80 MHz | 30 MHz                                               | 50 MHz | 100 MHz | 30 MHz                | 60 MHz | 100 MHz |                     |
| H1100NL <sup>3a</sup>            | 2:1         | 1:1 | -1.1                                          | -20                     | -14    | -11.5  | -42                                                  | -37    | -33     | -45                   | -40    | -35     | 1500                |
| H1101NL <sup>3a</sup>            | 1.25:1      | 1:1 | -1.1                                          | -20                     | -14    | -11.5  | -42                                                  | -37    | -33     | -45                   | -40    | -35     | 1500                |
| H1102NL <sup>3a</sup>            | 1:1         | 1:1 | -1.1                                          | -20                     | -14    | -11.5  | -42                                                  | -37    | -33     | -45                   | -40    | -35     | 1500                |
| H1121NL <sup>3c</sup>            | 1:1.414     | 1:1 | -1.1                                          | -18                     | -12    | -11.5  | -42                                                  | -37    | -30     | -45                   | -40    | -35     | 1500                |
| H1183NL <sup>3a</sup>            | 1:1         | 1:1 | -1.0                                          | -18                     | -14    | -12    | -40                                                  | -40    | -40     | -35                   | -35    | -35     | 1500                |
| H1199NL <sup>3a</sup>            | 1:1         | 1:1 | -0.9                                          | -18                     | -12    | -11    | -40                                                  | -35    | -30     | -45                   | -45    | -37     | 1500                |
| HX1188NL <sup>1, 3a</sup>        | 1:1         | 1:1 | -1.0                                          | -20                     | -14    | -12    | -42                                                  | -37    | -32     | -45                   | -40    | -35     | 1500                |
| HX1198 NL <sup>1, 3a</sup>       | 1:1         | 1:1 | -1.0                                          | -20                     | -14    | -12    | -42                                                  | -37    | -32     | -45                   | -40    | -35     | 1500                |
| H1302NL <sup>3a</sup>            | 1:1         | 1:1 | -1.1                                          | -16                     | -10    | -10    | -42                                                  | -37    | -32     | -35                   | -30    | -30     | 1500                |

- Operating temperature -40°C to +85°C
- For Tape & Reel packaging, add a "T" suffix at the end of the part number when ordering, e.g. H1102NLT.
- MSL = Moisture Sensitivity Level a=1 b=2 c=3 d=4

## Mechanical

### H11XXNL, H1121NL, \*H1183NL, H1199NL, HX1188NL, HX1198NL, H1302NL

\* For part number H1183NL, all dimensions are the same, except the height is .220" (5,59 mm).

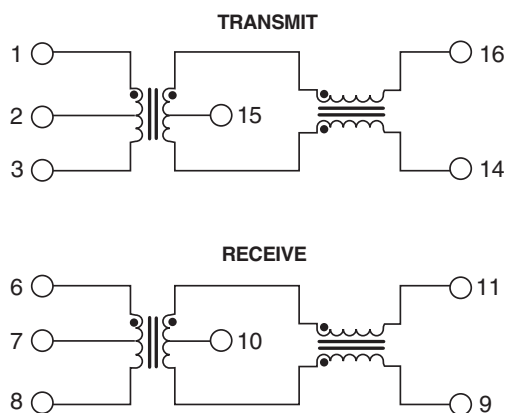


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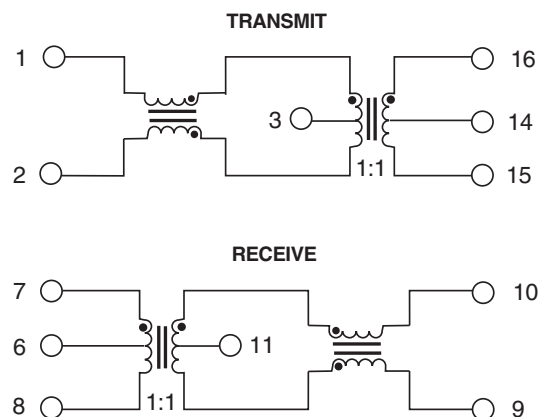


## Schematics

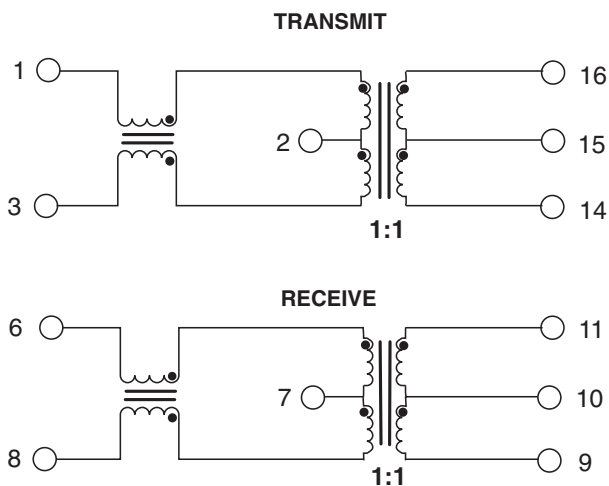
### H110XNL, H1121NL, HX1188NL



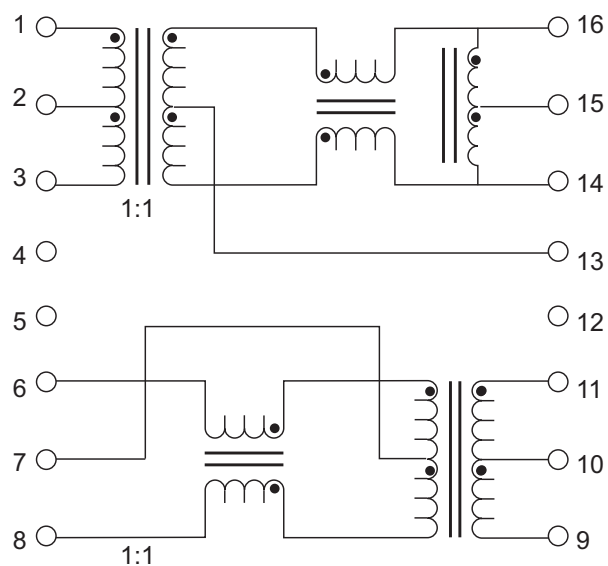
### H1199NL



### HX1198NL, H1183NL



### H1302NL

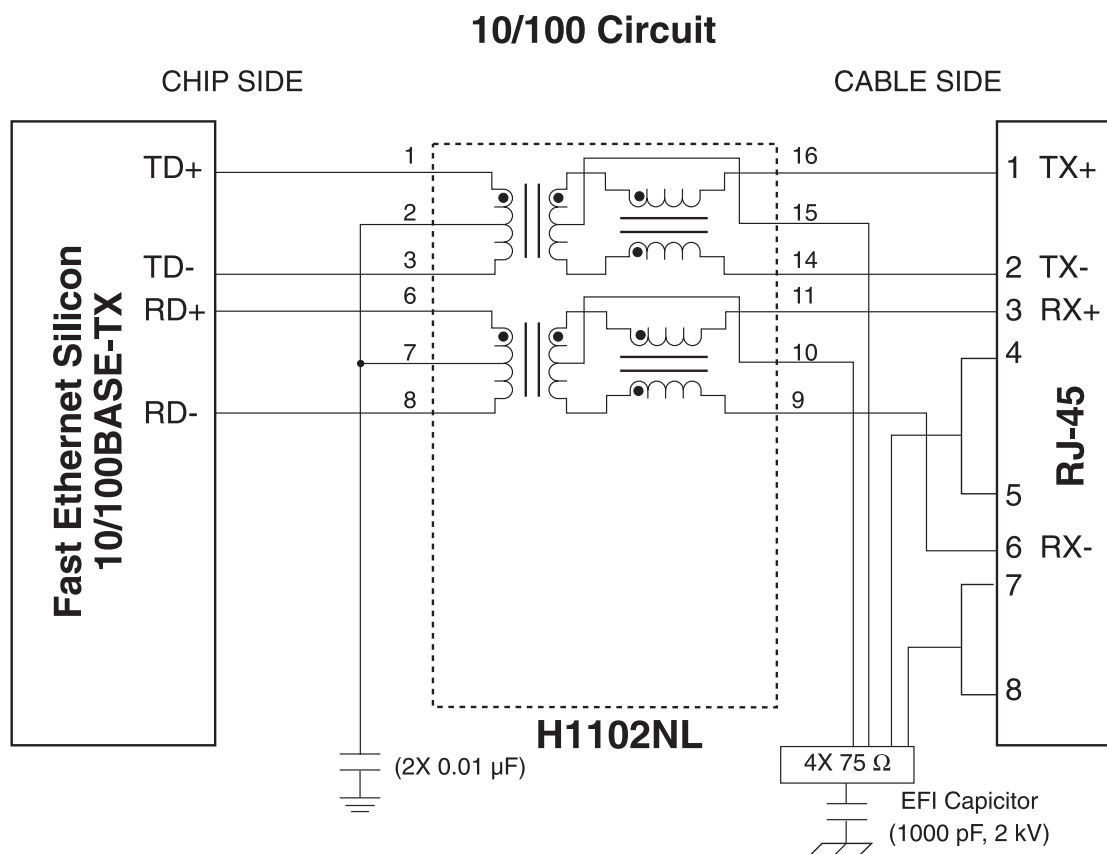


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## Typical Application Circuit

H1102NL



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